

A Theory of Controlled Propaganda for Decisive Warfare

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops a general theory of controlled propaganda as a strategic instrument of warfare. Drawing upon economics, finance, psychology, political science, philosophy, geopolitics, international relations, warfare economics, statistics, computer science, data science, and artificial intelligence, the paper formalizes propaganda as a state-controlled information allocation mechanism that reduces coordination costs, increases political cohesion, preserves military morale, and influences adversarial decision-making. We derive a set of propositions concerning credibility, morale, strategic signaling, alliance formation, and conflict outcomes. We demonstrate that propaganda functions as a force multiplier rather than an independent determinant of victory. The resulting framework provides a unified theory of information control in modern warfare.

The paper ends with “The End”

Contents

1 Introduction	2
2 Historical Foundations	2
3 Definition of Controlled Propaganda	3
4 Philosophical Foundations	3
5 Psychological Foundations	4
6 Political Economy of Propaganda	4
6.1 Coordination Cost Function	5
7 Economics of Wartime Information	5
8 Finance and Sovereign Stability	5
9 International Relations	6
10 Geopolitical Information Competition	6
11 The Credibility Constraint	7
12 Military Morale Dynamics	8
13 Statistical Framework	8
14 Machine Learning Perspective	8

15 AI-Driven Propaganda Systems	9
16 Algorithmic Framework	9
17 A General Theory of Decisive Warfare	10
18 Controlled Propaganda as a Force Multiplier	11
19 Illustrative Strategic Matrix	11
20 Conceptual Diagram	11
21 Conclusion	12

1 Introduction

Throughout history, warfare has involved not only the deployment of physical force but also the management of information.

Military campaigns require:

1. Political legitimacy.
2. Public support.
3. Economic mobilization.
4. Military cohesion.
5. Diplomatic alignment.
6. Strategic signaling.

Controlled propaganda refers to the deliberate management of information flows by a political authority for the purpose of influencing beliefs, expectations, incentives, and behavior.

The central claim of this paper is that propaganda is best understood as a coordination technology operating within a broader political-economic system.

2 Historical Foundations

The role of propaganda can be observed across:

1. Ancient empires.
2. Revolutionary movements.
3. Industrial warfare.
4. Cold War competition.
5. Contemporary information warfare.

Across these settings, successful propaganda systems share a common feature:

They amplify existing strengths rather than attempting to permanently conceal fundamental weaknesses.

3 Definition of Controlled Propaganda

Definition 1 (Controlled Propaganda). *Controlled propaganda is a centralized information allocation process designed to influence the beliefs and actions of targeted populations while maintaining sufficient credibility to preserve future influence.*

Let

$$P_t$$

denote propaganda intensity,

$$C_t$$

denote credibility,

$$M_t$$

denote morale,

and

$$W_t$$

denote wartime effectiveness.

We define

$$W_t = f(P_t, C_t, M_t).$$

4 Philosophical Foundations

The philosophy of propaganda lies at the intersection of epistemology and political legitimacy.

Governments seek to influence:

1. What citizens know.
2. What citizens believe.
3. What citizens expect others to believe.

This creates higher-order belief structures.

Let

$$B_i$$

denote citizen i 's belief.

Let

$$E(B_j)$$

represent citizen i 's expectation regarding citizen j 's belief.

Propaganda often targets

$$E(B_j)$$

rather than

$$B_j$$

directly.

This insight links propaganda to coordination theory and common knowledge problems.

5 Psychological Foundations

Propaganda exploits several psychological mechanisms:

1. Social proof.
2. Authority bias.
3. Repetition effects.
4. Identity reinforcement.
5. Fear management.
6. Hope generation.

Define morale as

$$M_t = \alpha S_t + \beta H_t - \gamma F_t,$$

where

- S_t is social cohesion,
- H_t is perceived probability of success,
- F_t is fear,
- $\alpha, \beta, \gamma > 0$.

Propaganda attempts to maximize M_t .

6 Political Economy of Propaganda

Consider a population of N citizens.

Each citizen chooses

$$s_i \in \{0, 1\},$$

where

$$s_i = 1$$

indicates support for wartime mobilization.

The probability of national mobilization is

$$Pr(Mobilization) = f\left(\sum_{i=1}^N E[s_i]\right).$$

Propaganda increases

$$E[s_i]$$

and therefore lowers coordination costs.

6.1 Coordination Cost Function

Let

$$K$$

denote coordination costs.

Then

$$K = \frac{\kappa}{1 + \lambda P},$$

where

$$\kappa, \lambda > 0.$$

Hence

$$\frac{\partial K}{\partial P} < 0.$$

Greater propaganda intensity reduces coordination costs.

7 Economics of Wartime Information

Consider a wartime production function

$$Y = AK^\alpha L^\beta M^\delta,$$

where

- K is capital,
- L is labor,
- M is morale,
- A is technology.

Then

$$\frac{\partial Y}{\partial M} > 0.$$

Propaganda indirectly raises output by increasing morale.

8 Finance and Sovereign Stability

Financial markets react to information.

Let

$$r_t$$

denote sovereign bond yields.

Suppose

$$r_t = r_f + \pi_t + \rho_t,$$

where

- r_f is the risk-free rate,
- π_t is expected inflation,
- ρ_t is political risk.

Controlled propaganda attempts to reduce

$$\rho_t.$$

Thus

$$\frac{\partial r_t}{\partial \rho_t} > 0.$$

A reduction in perceived political instability lowers borrowing costs.

9 International Relations

Propaganda targets five audiences:

1. Domestic civilians.
2. Domestic military.
3. Enemy civilians.
4. Enemy military.
5. Neutral actors.

Let

$$A_i$$

represent support from audience i .
Strategic influence becomes

$$I = \sum_{i=1}^5 w_i A_i,$$

where

$$w_i > 0.$$

10 Geopolitical Information Competition

Modern geopolitical competition occurs simultaneously across:

1. Military domains.
2. Economic domains.
3. Information domains.
4. Cyber domains.

Define total strategic power as

$$S = M + E + I + C,$$

where

- M is military capability,
- E is economic capability,
- I is information capability,
- C is cyber capability.

Propaganda primarily contributes to I .

11 The Credibility Constraint

The greatest limitation of propaganda is credibility.

Let

$$C_t$$

denote credibility.

Assume

$$0 \leq C_t \leq 1.$$

Let propaganda intensity be

$$P_t.$$

Excessive exaggeration produces

$$C_{t+1} = C_t - \eta P_t^2,$$

where

$$\eta > 0.$$

Thus credibility eventually declines when propaganda becomes detached from observable reality.

Theorem 1 (Credibility Constraint). *No propaganda system can indefinitely increase influence while credibility continuously declines.*

Proof. Influence can be written as

$$I_t = P_t C_t.$$

If

$$C_t \rightarrow 0,$$

then

$$I_t = P_t C_t \rightarrow 0$$

for any bounded propaganda intensity.

Therefore credibility collapse implies influence collapse.

□

12 Military Morale Dynamics

Define battlefield effectiveness

$$B_t.$$

Let

$$B_t = \theta T_t + \phi M_t,$$

where

- T_t denotes tactical competence,
- M_t denotes morale.

Then

$$\frac{\partial B_t}{\partial M_t} = \phi.$$

Thus morale serves as a force multiplier.

13 Statistical Framework

Suppose battlefield outcomes are represented by

$$Y_i = \beta_0 + \beta_1 M_i + \beta_2 E_i + \beta_3 T_i + \varepsilon_i.$$

Variables include:

- morale M_i ,
- economic capacity E_i ,
- military capability T_i .

Empirical estimation may be conducted using panel-data methods.

14 Machine Learning Perspective

Let

$$X$$

denote observable information variables.

A predictive model may estimate

$$Pr(\text{Victory}|X).$$

For example,

$$Pr(\text{Victory}) = \sigma(\beta' X),$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Features may include:

1. Information dominance.
2. Morale indicators.
3. Economic indicators.
4. Battlefield performance.
5. Alliance support.

15 AI-Driven Propaganda Systems

Modern systems increasingly employ AI.

Potential components include:

1. Sentiment analysis.
2. Narrative optimization.
3. Audience segmentation.
4. Reinforcement learning.
5. Information diffusion models.

Let

$$R_t$$

denote strategic reward.

An AI system solves

$$\max_{\pi} E \left[\sum_{t=0}^T \gamma^t R_t \right].$$

16 Algorithmic Framework

Pseudo-Code

Input:

Population data
Sentiment indicators
Battlefield indicators

Initialize:

Credibility C
Morale M

While war continues:

Measure sentiment

Estimate morale

Estimate credibility

Select narrative

Broadcast narrative

Update morale

Update credibility

End While

17 A General Theory of Decisive Warfare

We define national war power as

$$W = M + E + L + I,$$

where

- M is military power,
- E is economic power,
- L is legitimacy,
- I is information power.

Controlled propaganda primarily affects

$$L$$

and

$$I.$$

Proposition 1. *Propaganda is decisive only when military and economic balances are sufficiently close.*

Proof. Suppose two states possess war powers

$$W_A$$

and

$$W_B.$$

If

$$|M_A - M_B|$$

and

$$|E_A - E_B|$$

are extremely large,

then marginal improvements in legitimacy and information cannot reverse the outcome.

However, when military and economic capabilities are approximately equal, changes in legitimacy and information may alter

$$W_A - W_B.$$

Therefore propaganda is potentially decisive only near competitive margins.

□

18 Controlled Propaganda as a Force Multiplier

The central result of this paper is

$$Victory \neq Propaganda$$

but

$$Victory = Military + Economic + Political + Information.$$

Propaganda amplifies existing strengths.

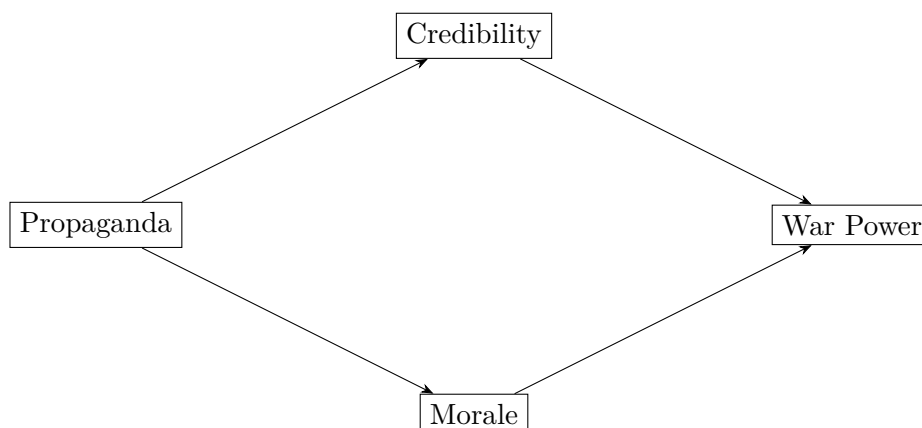
It does not permanently substitute for them.

19 Illustrative Strategic Matrix

Capability	Direct Effect	Indirect Effect	Strategic Role
Military	Combat Power	Deterrence	Core
Economic	Production	Sustainability	Core
Political	Legitimacy	Mobilization	Essential
Information	Morale	Coordination	Multiplier
Cyber	Disruption	Intelligence	Emerging

Table 1: Strategic Components of National Power

20 Conceptual Diagram



21 Conclusion

Controlled propaganda is best interpreted as an information-management technology that operates through credibility, morale, legitimacy, and coordination. Its principal function is to transform information into collective action. Although propaganda alone rarely determines conflict outcomes, it can become decisive when military and economic balances are close. The theory developed herein unifies insights from psychology, economics, political science, international relations, warfare studies, data science, and artificial intelligence into a coherent analytical framework for understanding information power in modern conflict.

References

- [1] Clausewitz, C. *On War*. Princeton University Press.
- [2] Schelling, T. *The Strategy of Conflict*. Harvard University Press.
- [3] Sun Tzu. *The Art of War*.
- [4] Downs, A. *An Economic Theory of Democracy*. Harper and Row.
- [5] Olson, M. *The Logic of Collective Action*. Harvard University Press.
- [6] Shannon, C. *A Mathematical Theory of Communication*. Bell System Technical Journal.
- [7] Kahneman, D. *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- [8] Tufekci, Z. *Twitter and Tear Gas*. Yale University Press.
- [9] Russell, S., and Norvig, P. *Artificial Intelligence: A Modern Approach*. Pearson.

Glossary

Controlled Propaganda Strategic management of information by a political authority.

Credibility Degree to which information is trusted.

Morale Psychological willingness to sustain effort.

Coordination Cost Cost of organizing collective action.

Information Power Ability to shape beliefs and expectations.

Political Legitimacy Perceived rightfulness of political authority.

Strategic Signaling Communication designed to influence adversarial expectations.

Information Warfare Competition through information rather than direct force.

Force Multiplier Factor that increases effectiveness of existing capabilities.

AI-Driven Propaganda Algorithmically optimized information dissemination.

The End